

Challenge 2 — Home Handyman Portfolio

COMP5450 Mobile Programming · Group 5

Overview

A Flutter portfolio app for a fictional home handyman service. The app has six screens — Home, Services, Portfolio, About, Reviews, and Contact — connected by a Material 3 bottom navigation bar. Data is served through a repository pattern using mock data, designed so Firebase Firestore can be swapped in later without changing any screen code.

App Description

- **Home** — Hero banner with quick-action buttons (Call, Book, Portfolio), stat chips showing years of experience, projects completed, and repeat client rate, plus a grid of featured services.
- **Services** — Two-column grid of all six service categories (Plumbing, Electrical, Carpentry, Painting, Tiling, General Repairs), each with a color-coded icon and description.
- **Portfolio** — Scrollable list of completed project cards with placeholder images, category tags, before/after labels, and short descriptions.
- **Reviews** — Average rating header and customer testimonial cards with name, date, star rating, and quoted comment.
- **Contact** — Business info (phone, email, hours, service area) and a message form with client-side validation.
- **About** — Accessible from the app bar info icon. Shows bio, certifications, and service area.

Architecture

The codebase follows a **repository pattern**. The abstract `PortfolioRepository` defines four methods: `getServices()`, `getProjects()`, `getReviews()`, and `submitContact()`. The current implementation, `MockPortfolioRepository`, returns data from `mock_data.dart`. Swapping to Firestore later requires only creating a new class that implements the same interface.

Project Structure

```
lib/  
main.dart  
models/  
service.dart, project.dart, review.dart  
data/  
portfolio_repository.dart (abstract)  
mock_portfolio_repository.dart  
mock_data.dart  
screens/  
home_screen.dart, services_screen.dart,  
portfolio_screen.dart, about_screen.dart,  
reviews_screen.dart, contact_screen.dart  
widgets/
```

```
hero_banner.dart, stat_chip.dart,  
service_card.dart, project_card.dart,  
review_card.dart
```

How to Configure and Run

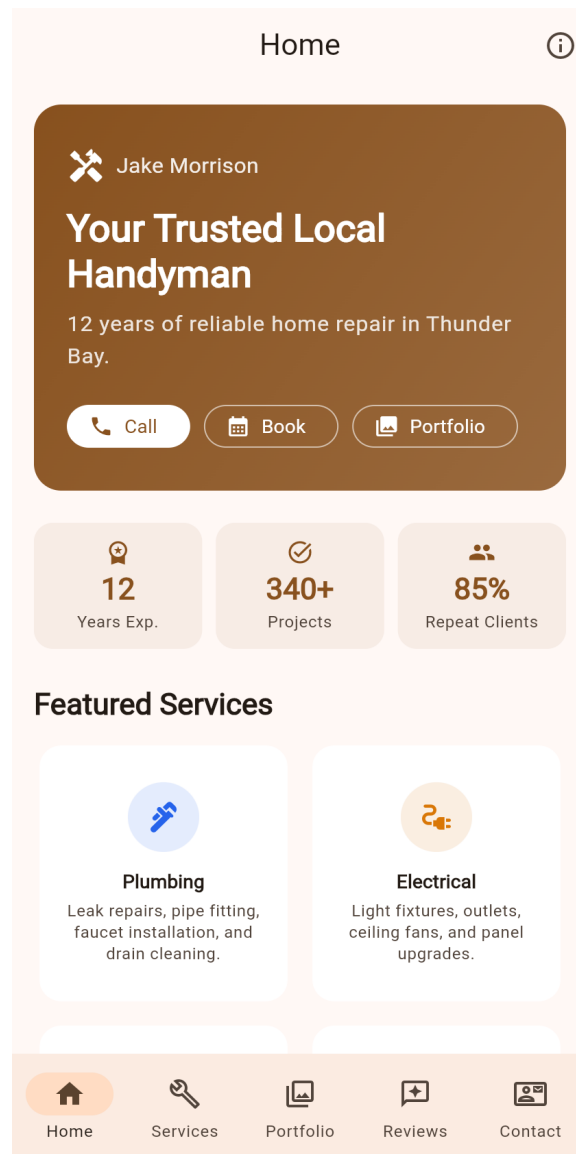
1. Install Flutter 3.32+ and add to PATH.
2. Clone the repo: `git clone https://github.com/lmc-eu/mobile_programming.git`
3. `cd challenges/challenge-2`
4. `flutter pub get`
5. `flutter run -d chrome` (or pick an Android emulator)

Testing

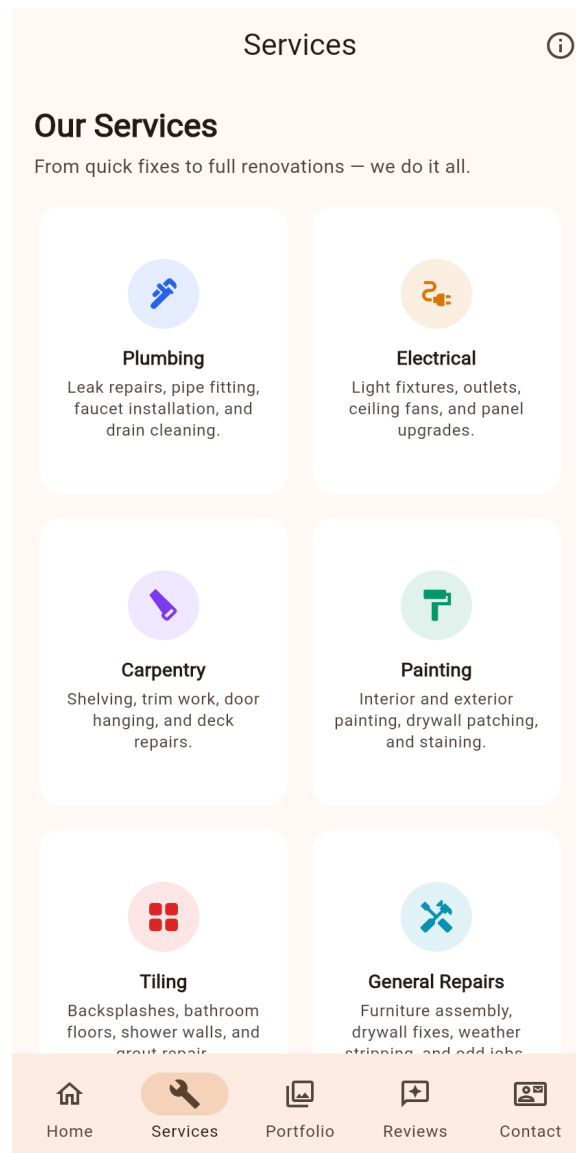
`flutter analyze` — zero issues. `flutter test` — 7 widget tests, all pass. Tests cover: app shell rendering, each bottom-nav tab, About page navigation, and contact form validation.

Screenshots

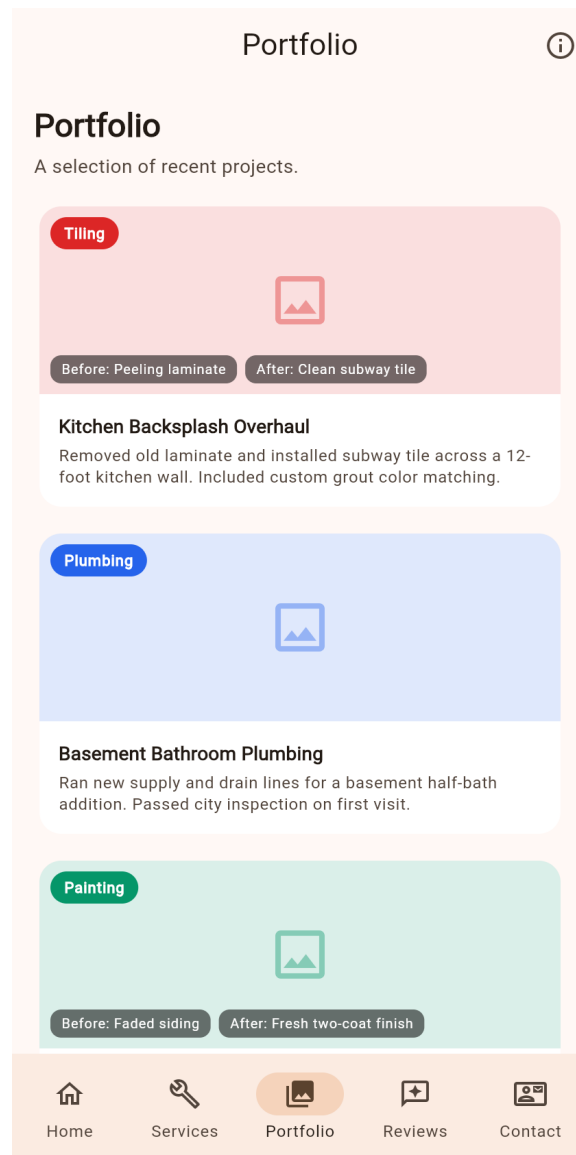
Home Screen



Services Screen



Portfolio Screen



GitHub Repository

<https://github.com/achenachena/handyman-portfolio>

Team — Group 5

(see D2L for member list)